

Laboratory Exercise 3.0: Configuring a Network with Two Routers

Objectives:

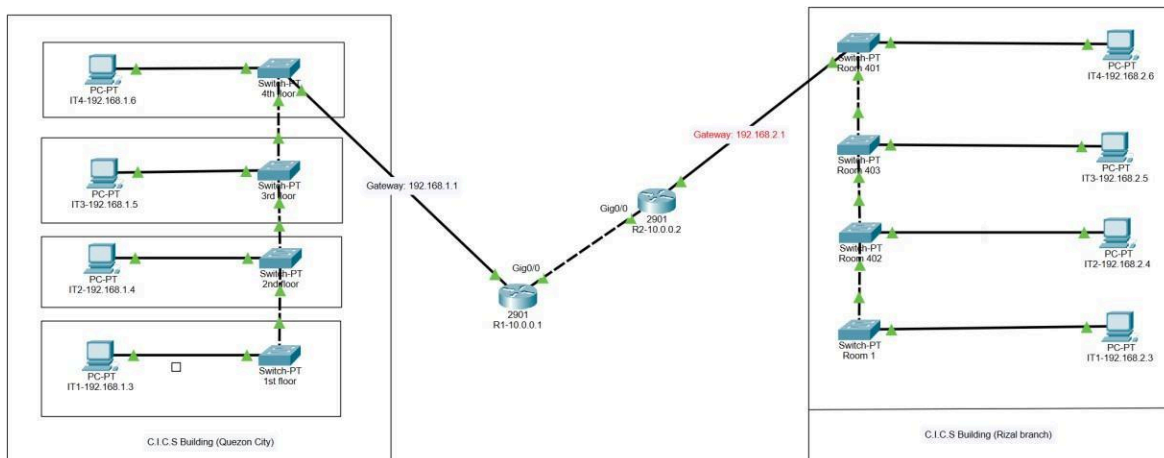
- Configure two routers to establish a network connection.
- Apply basic router configuration commands.
- Practice setting up and verifying network connectivity.

Equipment:

- Two routers (Cisco Packet Tracer)
- Network switches
- PCs with network adapters
- Ethernet cables
- Network configuration software (e.g., Cisco Packet Tracer)

Task 1: Network Topology and Planning

1. Draw a network diagram: Sketch a network topology using two routers, switches, and PCs. Label the connections and IP addresses for each device.
2. Define IP addressing scheme: Choose a subnet mask and assign IP addresses to routers, switches (if used), and PCs.
3. Plan router configuration: Configure the hostname, management interface IP address, and default gateway for each router.



Task 2: Router Configuration

1. Connect routers: Use the network configuration software or physical cables to connect the routers according to your diagram.
2. Configure basic settings: Set the hostname, management interface IP address, and subnet mask for each router.
3. Enable routing protocols: Depending on your chosen topology, configure RIP or OSPF routing protocols on the routers.

Task 3: Verification and Testing

1. Connect PCs and switches: Connect PCs to the network using switches or directly to the routers.
2. Assign IP addresses to PCs: Configure static IP addresses on the PCs according to your chosen addressing scheme.
3. Verify connectivity: Use ping commands to test connectivity between PCs and across the routers.
4. Troubleshoot and resolve connectivity issues: Identify and fix any configuration errors or connectivity issues.

Report:

- Document the network configuration for each router and switch.

Device Name	IP Address	Cable type	Origination Port	Destination Port	Status
R2	10.0.01	Copper Straight-through	R2-10.0.0.2:GigabitEthernet0/1	Room 401:FastEthernet2/1	Green

- Include screenshots or command outputs to illustrate key steps.
- Discuss the challenges encountered and solutions implemented.
- Evaluate the network performance and suggest improvements.